# Israel Souza Ferreira

## Machine Learning Engineer / Applied ML

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## Professional Summary

Computer Science graduate working on multimodal ML research, representations, and experimental LLM integration. Builds ML systems with APIs, vector search, and measured CPU performance.

## Professional Experience

### Tieta.ai · Machine Learning R&D · Remote

R&D Researcher: Jul 2026–present | ML Researcher, CNPq RHAE Fellow (part-time): Apr 2025–Jun 2026

Applied R&D in multimodal machine learning and representation systems for educational analysis.

- Contributed to developing and evaluating embeddings, vector representations, and similarity search for multimodal educational content.
- Collaborated on deep learning implementations and experiments combining text and image information and investigating representation alternatives.
- Contributed to experimental integration of LLMs into machine learning pipelines for structured information extraction and semantic representation enrichment.
- Supported applied research through scientific literature review, experiment implementation, and quantitative evaluation of modeling alternatives.

### Independent consulting · Educational institution · Mar 2026–present

Full Stack Developer and ML Engineer · Face Clock Evoluir

v1.3.0-rc.3: supervised institutional pilot candidate deployed to a test environment.

- Built 1:N facial identification, FastAPI APIs, and PostgreSQL/pgvector persistence for a CPU-only server; fixed concurrent extraction failures.
- Recalibrated the facial threshold: unknown-person acceptance fell from 45.4% to 0.2% in a controlled test. Calibration on the institution’s population remains pending.
- Used benchmarks to retain exact search and reduce frames; migrated document-classifier inference to ONNX Runtime with decision-parity checks on the held-out test.

## Research and Projects

### Triple Roman Domination in Graphs · Undergraduate thesis · 2024–2025

- Identified a counterexample to a published ILP formulation; implemented a corrected model, GA, and ACO (C++, Python/Pyomo). Evaluated 362 graphs; one run per instance limits conclusions.

## Technical Skills

Python, PyTorch, scikit-learn; embeddings, vector search, deep learning, LLM integration.

FastAPI, PostgreSQL/pgvector, Docker, ONNX Runtime, Git, pytest; SQL, Pandas, NumPy.

Cross-validation, calibration, ablation, classification metrics; C++, GA, ACO, Python/Pyomo.

## Education

### B.Sc. in Computer Science · Federal University of Ceará (UFC), Quixadá, Brazil

2021–19 Aug 2026 · Thesis defended in Feb 2025.

## Languages

Native Portuguese. English: good communication; fluent technical reading of papers and documentation.